

ABSTRACT REVIEW

Autism Spectrum Disorder Analysis for Toddlers

Batch No : B20

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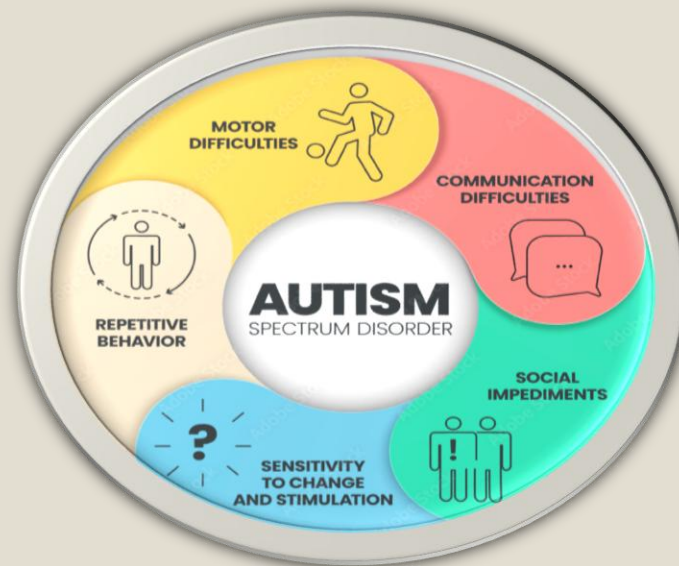
Under the Guidance of :-
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Abstract

- ❖ Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that impacts communication, social interaction, and behavior.
- ❖ Early detection is essential for improving outcomes through timely interventions.
- ❖ This project presents a web-based autism screening tool that uses responses to 10 key behavioral questions to assess ASD risk levels.
- ❖ A trained machine learning (ML) model classifies results into "Mild," "Severe," or "No ASD Traits" and provides relevant precautionary recommendations.

Introduction to Autism Detection

- ❖ This system begins by collecting basic information about the child, such as age and gender, followed by 10 carefully selected behavioral questions.
- ❖ Based on the responses, a trained machine learning model evaluates the risk of ASD and classifies it as "No ASD Traits," "Mild," or "Severe."
- ❖ The system then provides precautionary recommendations and possible treatment steps, offering parents and caregivers an accessible and effective tool for early autism screening.



Problem statement

Develop a machine learning-based system for early detection of ASD and recommendation of personalized treatment plans.

- ASD is a complex neurodevelopmental disorder affecting social interaction, communication, and behavior.
- Early intervention significantly improves long-term outcomes.
- Traditional diagnosis is often subjective, time-consuming, and reliant on specialist availability.

Literature survey

Author(s)	Tool(s) / Dataset(s)	Contribution / Method
Wall et al. (2012)	ADI-R, SVM	Reduced 93 ADI-R items to 7; achieved 99% accuracy using SVM.
Duda et al. (2016)	ADOS, ML classifiers	Used ADOS data for classification; achieved 96.8% accuracy.
Heinsfeld et al. (2018)	ABIDE (fMRI), Deep Learning (CNN)	Brain imaging-based detection; ~70% accuracy.
Hashemi et al. (2015)	3D facial imaging	Used facial morphology of toddlers to detect ASD.
Khalifa et al. (2021)	Speech, Eye-tracking data, ANN/CNN	Achieved >90% accuracy using deep learning.
UCI ASD Dataset	Behavioral questionnaire (M-CHAT, Q-CHAT)	Used in various ML models for severity classification.

Existing Systems

Autism detection has traditionally relied on standardized clinical tools such as the **Autism Diagnostic Observation Schedule (ADOS)** and the **Autism Diagnostic Interview – Revised (ADI-R)**.

These tools are comprehensive and widely accepted but require trained professionals and are time-consuming.

Disadvantages

Limited accessibility: Requires expert evaluation, which may not be available in remote areas.

Subjectivity: Depends on parental responses and clinical judgment, leading to inconsistencies.

Lack of instant feedback: Parents must wait for expert assessment and guidance.

Proposed system

- ❖ A web-based application that provides an automated autism screening test for toddlers.
- ❖ Uses a machine learning (ML) model to analyze responses from a 10-question test.
- ❖ Provides immediate classification results and treatment recommendations.

Advantages

- ❖ **Faster screening:** Instant analysis and severity classification.
- ❖ **Accessible:** Can be used by parents and caregivers from any location.
- ❖ **Objective results:** Uses ML for data-driven predictions.
- ❖ **Guidance & support:** Offers precautionary measures and possible treatment options.

Conclusion

Early detection of autism in toddlers is a collaborative effort between families and healthcare providers.

Accessing community resources and support groups can provide valuable assistance for families.

Continued research and awareness are vital for improving autism detection and intervention strategies.

